

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Department of Chemistry

CHY104-MATERIALS CHEMISTRY FOR DEVICES AND E-WASTE MANAGEMENT

MULTIPLE CHOICE QUESTIONS
UNIT – I ELECTRODE & ENERGY SYSTEMS

1.	In a galvanic cell,			
	A)	Anode is positive terminal	B)	Cathode is negative terminal
	C)	Anode is negative terminal	C)	Reaction is non-spontaneous
Ans	C)	Anode is negative terminal		
2.	An electrode is considered as a cathode when it has			
	A)	High reduction potential	B)	Low reduction potential
	C)	High oxidation potential	D)	Depends on the nature of metal
Ans	A)	High reduction potential		
3.	A working concentration cell is made of two half cells having			
	A)	Identical electrolytes and identical concentrations	B)	Identical electrolytes but different concentrations
	C)	Different electrolytes and identical concentrations	D)	Different electrolytes and different concentrations
Ans	B)	Identical electrolytes but different concentrations		
4.	Electrode potential of calomel electrode			
	A)	Varies directly with concentration of KCl	B)	Varies inversely with concentration of KCl
	C)	Is independent of concentration of KCl	D)	Depends only on the concentration of calomel
Ans	B)	Varies inversely with concentration of KCl		
5.	When clubbed with copper electrode, calomel electrode acts as			
	A)	Anode	B)	Cathode
	C)	Either as anode or cathode	D)	Auxiliary electrode
Ans	A)	Anode		
6.	A calomel electrode is an example for			
	A)	Reference electrode	B)	Ion selective electrode
	C)	Inert electrode	D)	Gas electrode
Ans	A)	Reference electrode		
7.	The calomel electrode is represented as			
	A)	Hg Hg ₂ Cl ₂ (s) KCl (saturated or standard)	B)	Ag AgCl (s) KCl (saturated)
	C)	Hg HgCl ₂ (s) KCl (saturated or standard)	D)	Ag AgCl (s) HCl (0.1M) glass
Ans	A)	Hg Hg ₂ Cl ₂ (s) KCl (saturated or standard)		
8.	Which is the sparingly soluble salt present in calomel electrode?			
	A)	HgCl	B)	Hg ₂ Cl ₂
	C)	KCl	D)	HgCl ₂
Ans	B)	Hg ₂ Cl ₂		
9.	EMF of electrolyte concentration cell is independent of			

	A)	Number of electrons involved	B)	Temperature
	C)	Standard electrode potential of electrode	D)	Electrolyte concentration
Ans	C)	Standard electrode potential of electrode		
10.	The potential of the calomel electrode depends on the concentration of			
	A)	Hg	B)	Hg ₂ Cl ₂
	C)	KOH	D)	KCl
Ans	D)	KCl		
11.	What is the electrode potential of the calomel electrode with saturated KCl solution at 298K?			
	A)	0.334 V	B)	0.281 V
	C)	0.242 V	D)	0.0591 V
Ans	C)	0.242 V		
12.	The reference electrode that can be used to determine the potential of zinc electrode is			
	A)	Glass electrode	B)	Platinum electrode
	C)	Copper electrode	D)	Calomel electrode
Ans	D)	Calomel electrode		
13.	A glass electrode is commonly used to measure			
	A)	Oxidation potential	B)	Reduction potential
	C)	pH of a solution	D)	Conductivity of a solution
Ans	C)	pH of a solution		
14.	Glass electrode is sensitive to ions.			
	A)	Sodium	B)	Hydrogen
	C)	Chloride	D)	Hydroxide
Ans	B)	Hydrogen		
15.	Which of the following is an example for ion selective electrode?			
	A)	Platinum electrode	B)	Gas electrode
	C)	Calomel electrode	D)	Glass electrode
Ans	D)	Glass electrode		
16.	Identify which of the following statement/s does/do not hold true for glass electrode? (i) The glass membrane is made of a mixture of silicon, sodium and carbon oxides. (ii) The glass membrane exhibits electrical conductivity. (iii) The potential of the glass electrode is a combination of boundary potential, asymmetry potential and internal reference electrode potential.			
	A)	Only (i)	B)	Only (ii)
	C)	Both (i) and (iii)	D)	Both (ii) and (iii)
Ans	A)	Only (i)		
17.	Which of the following is an example for working concentration cell?			
	A)	Fe FeSO ₄ (0.10M) CuSO ₄ (0.10M) Cu	B)	Mg Mg ²⁺ (aq) Cd ²⁺ (aq) Cd
	C)	Ag AgNO ₃ (0.10M) AgNO ₃ (0.50M) Ag	D)	Fe Fe ²⁺ (0.015M) Ag ⁺ (0.13M) Ag
Ans	C)	Ag AgNO ₃ (0.10M) AgNO ₃ (0.50M) Ag		
18.	Among the below given statements, which of them are true for concentration cell? (i) Electrode immersed in dilute solution behaves as anode. (ii) The Nernst equation of a concentration cell is independent of temperature and electrode. (iii) Concentration cells can be used to measure ion concentration in a solution.			
	A)	(i), (ii) and (iii)	B)	(i) and (ii)
	C)	(ii) and (iii)	D)	(i) and (iii)

Ans	D)	(i) and (iii)		
19.	The EMF of the concentration cell $\text{Ag(s)} \text{Ag}^+(0.05\text{M}) \text{Ag}^+(0.05\text{M}) \text{Ag}$ at 298°K is			
	A)	0.0591V	B)	0.02955V
	C)	0V	D)	-0.0591V
Ans	C)	0V		
20.	The number of galvanic cells, each of 1.1 V, required to construct a 11 V battery is			
	A)	15	B)	22
	C)	11	D)	10
Ans	D)	10		

21.	Which of the following is not a component of a battery?			
	A)	Cathode	B)	Salt-bridge
	C)	Anode	D)	Electrolyte
Ans	B)	Salt-bridge		
22.	Which of the following is not a feature of secondary batteries?			
	A)	Rechargeable	B)	Long-lasting when charged
	C)	Non-rechargeable	D)	Can be used after recharging
Ans	C)	Non-rechargeable		
23.	 is used as anode material in commercial lithium-ion batteries.			
	A)	Lithium intercalated in Graphite layers	B)	Silicon
	C)	Lithiated metal oxides	D)	Platinum
Ans	A)	Lithium intercalated in Graphite layers		
24.	The role of polypropylene separator in lithium-ion battery is to			
	A)	Block the flow of Ions	B)	Allow the free flow of ions
	C)	Allow the free flow of electrons	D)	Block the flow of electrons
Ans	B)	Allow the free flow of ions		
25.	In lithium-ion battery, the electrolyte used is			
	A)	Polar solvent	B)	Non-polar solvent
	C)	Lithium salts dissolved in polar solvent	D)	Lithium salts dissolved in non-polar solvent
Ans	D)	Lithium salts dissolved in non-polar solvent		
26.	Special properties of 'Li' metal that make it advantageous as an electrode material include			
	A)	Light weight	B)	Low electrical conductivity
	C)	High electrode potential	D)	Low cost
Ans	A)	Light weight		
27.	The process during charging of a Li-ion battery involves			
	A)	Li ⁺ ions moving from cathode to anode		Li ⁺ ions moving from anode to cathode
	C)	H ⁺ moving from cathode to anode		H ⁺ moving from anode to cathode
Ans	A)	Li ⁺ ions moving from cathode to anode		
28.	In flow batteries, the electro-active material is stored in			
	A)	Electrode	B)	Electrolyte
	C)	Separator	D)	Membrane
Ans	B)	Electrolyte		
29.	Identify the incorrect statement for vanadium redox flow battery.			
	A)	The redox reactions involve vanadium in different oxidation states.	B)	The electrolytes are stored externally and pumped through the cell.

	C)	The energy is stored in the electrode material.	D)	The electrolytes are rechargeable.
Ans	C)	The energy is stored in the electrode material.		
30.	In a vanadium redox flow battery, the electrolyte contains			
	A)	Vanadium ions of same oxidation states	B)	Vanadium ions in different oxidation states
	C)	Lithium ions	D)	Copper ions
Ans	B)	Vanadium ions in different oxidation states		
31.	Which ion pair is used in the negative half-cell of a vanadium redox flow battery?			
	A)	V^{3+}/V^{2+}	B)	V^{5+}/V^{4+}
	C)	Fe^{3+}/Fe^{2+}	D)	Cu^{2+}/Cu^{+}
Ans	A)	V^{3+}/V^{2+}		
32.	In VRFBs, the battery energy and capacity depend on			
	A)	Electrode material only	B)	The volume and concentration of electrolytes
	C)	The size of the membrane	D)	The temperature of operation
Ans	B)	The volume and concentration of electrolytes		
33.	Photovoltaic cells convert energy to energy.			
	A)	Chemical; Electrical	B)	Electrical; Chemical
	C)	Solar; Electrical	D)	Electrical; Solar
Ans	C)	Solar; Electrical		
34.	The most commonly used semiconductor material in commercial solar cells is			
	A)	Gallium arsenide (GaAs)	B)	Silicon (Si)
	C)	Cadmium sulfide (CdS)	C)	Germanium (Ge)
Ans	B)	Silicon (Si)		
35.	The working principle of PV cell is based on			
	A)	Ohm's law	B)	Nernst Equation
	C)	Newtons laws	D)	Photoelectric effect
Ans	D)	Photoelectric effect		
36.	Which of the following statement/s is/are true?			
	(i) In a PV cell, the p-doped silicon layer is exposed to sunlight.			
	(ii) Depletion zone is responsible for separation of electrons and holes.			
	A)	Only (i)	B)	Only (ii)
	C)	Both (i) and (ii)	D)	Neither (i) nor (ii)
Ans	B)	Only (ii)		
37.	Which of the following property makes it easy to purify silane and chlorosilanes?			
	A)	High density	B)	High boiling point
	C)	Volatility	D)	Magnetic property
Ans	C)	Volatility		
38.	Which compound is formed when silane is pyrolyzed at 600–900°C?			
	A)	SiO ₂	B)	Silicon nitride
	C)	Trichlorosilane	D)	Polysilicon
Ans	D)	Polysilicon		

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MULTIPLE CHOICE QUESTIONS
UNIT – II CORROSION SCIENCE & MULTIPLE FINISHING

1.	Identify the incorrect statement/s.			
	(i) Corrosion is a natural process initiating on the surface of the metal.			
	(ii) Corrosion reduces metal to its ionic form.			
	(iii) Rate of corrosion is independent of the environment.			
	A)	(ii) and (iii)	B)	(i) and (ii)
	C)	Only (iii)	D)	Only (i)
Ans	A)	(ii) and (iii)		
2.	Situation favorable for dry corrosion is			
	A)	Presence of moisture	B)	Presence of conducting electrolyte
	C)	Presence of oxygen or other gases	D)	Presence of non-polar liquids
Ans	C)	Presence of oxygen or other gases		
3.	Reduction of water / moisture in presence of oxygen leads to the formation of			
	A)	Hydrogen gas only	B)	Hydroxyl ions only
	C)	Both hydrogen gas and hydroxyl ions	D)	Oxide anions
Ans	B)	Hydroxyl ions only		
4.	Corrosion that takes place when two dissimilar metals are coupled and exposed to the same environment, is known as			
	A)	Galvanic corrosion	B)	Dry corrosion
	C)	Pitting corrosion	D)	Passivation
Ans	A)	Galvanic corrosion		
5.	Which of the following statements holds true for Galvanic Corrosion?			
	A)	More cathodic metal gets corroded.	B)	Less noble metal gets corroded.
	C)	The metal having a higher standard reduction potential gets corroded.	D)	The metal placed lower in the galvanic series get corroded.
Ans	B)	Less noble metal gets corroded.		
6.	A steel tank fitted with a copper tap undergoes corrosion in water because			
	A)	Electrode potential of both metals are almost equal	B)	Steel acts as an anode and corrodes due to galvanic action
	C)	Steel is susceptible to water	D)	Copper acts as an anode and corrodes
Ans	B)	Steel acts as an anode and corrodes due to galvanic action		
7.	Complete the statement: In differential aeration corrosion,			
	A)	Oxygen-deficient part gets protected.	B)	Oxygen-deficient part acts as anode.
	C)	Oxygen-rich part acts as anode.	D)	Metal as a whole acts as cathode.
Ans	B)	Oxygen-deficient part acts as anode		
8.	The accelerated corrosion of iron under defective tin coatings is due to			

	A)	Change in electrode potential	B)	Acidic environment
	C)	Small cathodic surface area	D)	Small anodic surface area
Ans	D)	Small anodic surface area		
9.	For the accelerated corrosion to take place ratio of anodic to cathodic area should be			
	A)	Greater than 1	B)	Lesser than 1
	C)	Equal to 1	D)	Greater than 1.5
Ans	B)	Lesser than 1		
10.	The metal, which is protected by a layer of its own oxide, is			
	A)	Al	B)	Pt
	C)	Zn	D)	Fe
Ans	A)	Al		
11.	In general, corrosion is maximum when the pH of the corroding medium is			
	A)	Above 7.0	B)	Equal to 7.0
	C)	Below 7.0	C)	Equal to 10.0
Ans	C)	Below 7.0		
12.	Identify the correct statement/s: (i) Lower the hydrogen overvoltage, more is the rate of corrosion. (ii) Smaller the anodic area (larger cathodic area), more is the rate of corrosion.			
	A)	Only (i)	B)	Only (ii)
	C)	Both (i) and (ii)	D)	Neither (i) nor (ii)
Ans	C)	Both (i) and (ii)		
13.	Iron corrodes faster in			
	A)	Hard water	B)	Soft water
	C)	Demineralised water	D)	Distilled water
Ans	A)	Hard water		
14.	CPR is expressed in unit.			
	A)	mg per year	B)	mm per year
	C)	mg per hr	D)	mm per hr
Ans	B)	mm per year		
15.	What is the role of NH_4Cl flux during galvanization process?			
	A)	Prevents oxidation of Zn	B)	Prevents oxidation of iron
	C)	Helps to maintain constant bath temperature	D)	Prevents chemical reaction between Zn and Fe
Ans	A)	Prevents oxidation of Zn		
16.	In anodic coatings, the coating metals possess reduction potential than base metal.			
	A)	Higher	B)	medium
	C)	Lower	D)	equal
Ans	C)	Lower		
17.	Anodic metal coating is a process of coating the base metal with			
	A)	more reactive metal	B)	inert metal
	C)	less reactive metal	D)	passive metal
Ans	A)	more reactive metal		
18.	Chemical conversion coating is type of coating.			
	A)	Organic	B)	Inorganic
	C)	Polymer	D)	Alloy
Ans	B)	Inorganic		

19.	Anodization of corrosion control technique is based on the principle of			
	A)	Chemical coating	B)	Chemical conversion coating
	C)	Metal coating	D)	Electroless plating
Ans	B)	Chemical conversion coating		
20.	Sacrificial anode method is widely used in ships and underground pipelines as			
	A)	It is highly decorative.	B)	It ensures uniform strength of the material.
	C)	It continuously protects the base metal by corroding the anodic metal.	D)	It eliminates the use of anodic metal.
Ans	C)	It continuously protects the base metal by corroding the anodic metal.		
21.	Sacrificial anodes used in cathodic protection of iron are generally made of			
	A)	Copper, Nickel, Lead	B)	Magnesium, Zinc, Aluminium
	C)	Carbon, Platinum, Gold	D)	Iron, Chromium, Silver
Ans	B)	Magnesium, Zinc, Aluminium		
22.	To protect the buried pipeline from corrosion, it is connected to Mg piece through a wire. This process is called as			
	A)	Impressed current cathodic protection	B)	Galvanic protection
	C)	Sacrificial cathodic protection	D)	Sacrificial anodic protection
Ans	D)	Sacrificial anodic protection		
23.	Which of the following is not a corrosion prevention technique?			
	A)	Sacrificial Anode	B)	Protective coating
	C)	Pickling	D)	Anodizing
Ans	C)	Pickling		

24.	In the electroplating process, generally the object to be plated is made as			
	A)	Anode	B)	Electrolyte
	C)	Cathode	D)	Power house
Ans	C)	Cathode		
25.	In electrodeposition process, complexing agents are added to			
	A)	increase the chemical reaction between the cathode metal and plating ions	B)	make the potential of the plating ion more negative
	C)	enhance the passivation of anodes	D)	decrease the throwing power of the plating bath
Ans	B)	make the potential of the plating ion more negative		
26.	The electrolyte commonly used for chromium electroplating is			
	A)	H ₂ SO ₄	B)	CrO ₃ + H ₂ SO ₄
	C)	CrO ₃ + HCl	D)	HCl solution
Ans	B)	CrO ₃ + H ₂ SO ₄		
27.	During chromium plating, is used as anode.			
	A)	Chromium	B)	Nickel
	C)	Calomel electrode	D)	Inert electrode
Ans	D)	Inert electrode		
28.	Chromium anodes are not used in Cr plating because			
	A)	Chromium metal passivates strongly in acid sulphate medium.	B)	Chromium metal activates strongly in acid sulphate medium.
	C)	Chromium anode gives rise to Cr ⁶⁺ ions on dissolution.	D)	Chromium metal oxidizes the base component

Ans	A)	Chromium metal passivates strongly in acid sulphate medium.		
29.	Electroless plating is characterized by			
	A)	The use of an external electrical current.	B)	A chemical reduction process without external current.
	C)	Electrodeposition of metals.	D)	The need for a conductive substrate.
Ans	B)	A chemical reduction process without external current.		
30.	Identify the advantage of electroless plating over electroplating technique.			
	A)	Higher deposition rates.	B)	Better throwing power of the bath solution.
	C)	Lower cost.	D)	Easier to control the deposition process.
Ans	B)	Better throwing power of the bath solution.		
31.	During the fabrication of PCB, the electroless plating of copper is performed using as reducing agent.			
	A)	Sodium hypophosphate	B)	Sodium borohydride
	C)	Hydrazine	D)	Formaldehyde
Ans	D)	Formaldehyde		
32.	In the production of double-sided printed circuit boards, connection-through-hole is efficiently made by			
	A)	Electroplating	B)	Electroless plating
	C)	Molten dip method	D)	using copper wires
Ans	B)	Electroless plating		
33.	Which of the following statements is correct regarding the difference between electroless and electrolytic plating?			
	A)	Electroless plating includes reduction reaction, while electrolytic plating does not.	B)	Electrolytic plating is a chemical process, while electroless plating is an electrochemical process.
	C)	Electroless plating offers more uniform coatings than electrolytic plating.	D)	Electrolytic plating is used for through-hole plating, while electroless plating is not
Ans	C)	Electroless plating offers more uniform coatings than electrolytic plating.		
34.	Electroless plating is used in the electronics industry for			
	A)	Creating decorative finishes on electronic components.	B)	Increasing the electrical resistance of circuits.
	C)	Creating conductive pathways on printed circuit boards (PCBs).	D)	Providing corrosion protection for metal components.
Ans	C)	Creating conductive pathways on printed circuit boards (PCBs).		
35.	In electronics, the abbreviation PCB stands for			
	A)	Polished Circuit Board	B)	Printed Circular Board
	C)	Polymer Circuit Board	D)	Printed Circuit Board
Ans	D)	Printed Circuit Board		

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MULTIPLE CHOICE QUESTIONS

UNIT – III ANALYTICAL TECHNIQUES, MEMORY DEVICES AND DISPLAY SYSTEMS

1.	Conductometry is an electrochemical method of analysis based on measuring the of an electrolytic solution.			
	A)	Resistance	B)	Dilution
	C)	Absorbance	D)	Viscosity
Ans	A)	Resistance		
2.	In conductivity cell, cell constant is the ratio of			
	A)	ℓ/a	B)	a/ℓ
	C)	ℓ/a^2	D)	a^2/ℓ
Ans	A)	ℓ/a		
3.	 is defined as the conductance of a solution containing one mole of an electrolyte.			
	A)	Specific conductance	B)	Equivalent conductance
	C)	Molar conductance	D)	Ionic conductance
Ans	C)	Molar conductance		
4.	In conductometric titrations of strong acid and strong base, the conductance			
	A)	First decreases steeply, then rises rapidly	B)	First rises rapidly, then decreases steeply
	C)	Remains same throughout titration	D)	Increases gradually
Ans	A)	First decreases steeply then rises rapidly		
5.	Conductance of an electrolyte is directly proportional to			
	A)	Mobility and concentration of ions	B)	Resistance of the electrolyte
	C)	Voltage applied across the electrolyte	D)	Temperature of the electrolyte
Ans	A)	Mobility and concentration of ions		
6.	During conductometric titration of mixture of strong and weak acid against NaOH, which of the following is true?			
	A)	Weak acid will be neutralized first	B)	Strong acid will be neutralized first
	C)	It depends on relative volumes of strong and weak acid taken	D)	It depends on relative concentrations of strong and weak acid taken
Ans	B)	Strong acid will be neutralized first		
7.	In conductometric titrations, the equivalence point is determined graphically by plotting a graph of			
	A)	Conductance vs Volume of alkali	B)	Absorbance vs Concentration
	C)	Conductance vs Acid concentration	D)	Conductance vs EMF
Ans	A)	Conductance vs Volume of alkali		
8.	At half-equivalence point, pH of a weak acid is considered equal to its pKa, because at that point			
	A)	The acid gets completely neutralized.	B)	The concentrations of salt and acid become equal
	C)	The concentration of the acid is maximum	D)	The salt concentration is maximum

Ans	B)	The concentrations of salt and acid become equal	
9.	In potentiometric titration of FAS against standard $K_2Cr_2O_7$, sharp increase in potential is observed when:		
	A)	All ferric ions are converted in to ferrous ions	B) 50% of ferric ions are converted in to ferrous ions
	C)	All ferrous ions are converted into ferric ions	D) 50% of ferric ions are converted in to ferrous ions
Ans	C)	All ferrous ions are converted into ferric ions	
10.	Potentiometric titration involves measurement of		
	A)	The potential between indicator electrode and reference electrode	B) The resistance of an electrolytic solution
	C)	pH of the solution	D) Absorbance of the solution
Ans	A)	The potential between indicator electrode and reference electrode	
11.	Identify the correct statement/s: (i) Calomel electrode is used as reference electrode. (ii) Platinum electrode is used to measure the conductance of a solution. (iii) Glass electrode is used to measure the potential of a solution.		
	A)	Only (i)	B) (ii) and (iii)
	C)	(i) and (iii)	D) (i), (ii) and (iii)
	Ans	A)	Only (i)
12.	Deviation from Beer's Lambert law is observed at		
	A)	Higher concentrations of analytes	B) Lower concentrations of analytes
	C)	Colorless analytes	D) Colored analytes
Ans	A)	Higher concentrations of analytes	
13.	Colorimetric determination of metal from its solution is dependent on		
	A)	Boyle's law	B) Planck's law
	C)	Charle's law	D) Beer-Lambert law
Ans	D)	Beer-Lambert law	
14.	In the colorimetric estimation of iron, which reagent is used to form a red-colored complex with Fe^{3+} ?		
	A)	Ammonia solution	B) Potassium thiocyanate
	C)	Acid solution	D) Copper sulphate
Ans	B)	Potassium thiocyanate	
15.	Identify the incorrect statement: (i) Copper-ammonia complex shows a maximum absorbance at 500 nm. (ii) Copper-ammonia complex has an intense blue color.		
	A)	Only (i)	B) Only (ii)
	C)	Both (i) and (ii)	D) Neither (i) nor (ii)
	Ans	A)	Only (i)
16.	Compared to chemical analysis, instrumental analysis		
	A)	Are much slower	B) Are much faster
	C)	Requires large amount of analyte	D) Are less useful
Ans	B)	Are much faster	
17.	Compared to chemical analysis, instrumental analysis requires		
	A)	Very large amount of analyte	B) Very small amount of analyte
	C)	No analyte	D) No calibration of instruments
Ans	B)	Very small amount of analyte	
18.	In an electrochemical sensor, the active component that converts chemical reactions into measurable electrical signals, is called a/an		

	A)	Receptor	B)	Amplifier
	C)	Detector	D)	Transducer
Ans	D)	Transducer		
19.	In industrial settings, oxides of nitrogen and sulphur are commonly detected using			
	A)	Disposable sensors	B)	Electrochemical sensors
	C)	Optical sensors	D)	Thermometric sensors
Ans	B)	Electrochemical sensors		
20.	What is the key feature of disposable sensors?			
	A)	High reusability	B)	Meant for single-use applications
	C)	Variable accuracy	D)	Does not suffer from deactivation
Ans	B)	Meant for single-use applications		

21.	For a material to be suitable for application in memory device, it must not			
	A)	Switch rapidly between states	B)	Be sensitive to small voltage changes
	C)	Have unstable states	D)	Store data for prolonged period
Ans	C)	Have unstable states		
22.	A PNP junction is normally found in electronic memory device.			
	A)	Transistor-type	B)	Capacitor-type
	C)	Resistor-type	D)	Charge-transfer-type
Ans	A)	Transistor-type		
23.	Number of terminals needed to construct transistor type electronic memory devices are			
	A)	1	B)	3
	C)	6	D)	2
Ans	B)	3		
24.	The charge-transfer effects observed in the memory device is due to the binding between molecules.			
	A)	Donor and Acceptor	B)	Cationic and Anionic
	C)	Diatomic and Polyatomic	D)	Elemental and Macro
Ans	A)	Donor and Acceptor		
25.	Identify the incorrect statement.			
	A)	Pentacene is an acene derivative.	B)	Pentacene structure consists of five fused benzene rings.
	C)	Pentacene exhibits very low charge carrier mobility.	D)	Perfluoropentacene behaves as n-type semiconductor.
Ans	C)	Pentacene exhibits very low charge carrier mobility.		
26.	An example of an organic molecule, which can replace 'n' type semiconductor, is			
	A)	Benzene	B)	Pentacene
	C)	Perfluoropentacene	D)	Fluro benzene
Ans	C)	Perfluoropentacene		
27.	What is the major advantage of organic electronic memory devices?			
	A)	High-operating voltage	B)	Excellent mechanical rigidity
	C)	Flexibility and low-cost fabrication	D)	Better performance at higher temperature
Ans	C)	Flexibility and low-cost fabrication		
28.	Which property of polymers makes them suitable for applications in memory devices?			
	A)	High electrical resistance	B)	Low thermal stability
	C)	Bistable state	D)	Brittle nature
Ans	C)	Bistable state		

29.	Which of the following structural feature is not a necessity for a molecule to exhibit liquid crystalline property?			
	A)	Rigid Core	B)	Heterocyclic moiety
	C)	Flexible end group	D)	Bridge group
Ans	B)	Heterocyclic moiety		
30.	The orientational behaviour of lyotropic crystals is a function of.....			
	A)	Concentration	B)	Solvent
	C)	Concentration and solvent	D)	Temperature
Ans	C)	Concentration and solvent		
31.	Nematic type of liquid crystal possess			
	A)	Positional order	B)	Directional order
	C)	Both positional and directional order	D)	Neither positional nor directional order
Ans	B)	Directional order		
32.	Which phase of liquid crystals constitute molecules having twisted helical structure?			
	A)	Nematic	B)	Isotropic
	C)	Smectic	D)	Cholesteric
Ans	D)	Cholesteric		
33.	Which liquid crystal shows layered molecular arrangement?			
	A)	Nematic	B)	Twisted Nematic
	C)	Smectic	D)	Discotic
Ans	C)	Smectic		
34.	Which liquid crystal is most commonly employed in LCD technology?			
	A)	Nematic	B)	Twisted Nematic
	C)	Smectic	D)	Discotic
Ans	B)	Twisted Nematic		
35.	In LCD units, liquid crystals act as wave guide by tilting the plane of polarized light by			
	A)	45°	B)	60°
	C)	90°	D)	120°
Ans	C)	90°		
36.	Identify the correct statement/s:			
	(i) In basic LCD, two polarizers are used which are oriented perpendicular to each other.			
	(ii) The role of the polarizers is to allow/block light waves of specific orientation.			
	A)	Only (i)	B)	Only (ii)
	C)	Both (i) and (ii)	D)	Neither (i) nor (ii)
	Ans	C)	Both (i) and (ii)	
37.	A numeric liquid crystal display has segments.			
	A)	Thirty-two	B)	Sixteen
	C)	Eight	D)	Seven
Ans	D)	Seven		
38.	Working principle of LED is			
	A)	Photo-luminescence	B)	Chemi-luminescence
	C)	Electrooptic effect	D)	Electro-luminescence
Ans	D)	Electro-luminescence		
39.	In organic light-emitting diodes, organic molecules replace			
	A)	The anode	B)	The cathode
	C)	Both anode and cathode	D)	Semiconducting materials
Ans	D)	Semiconducting materials		

40.	The color emitted by a quantum dot depends mainly on			
	A)	Applied voltage	B)	Operating temperature
	C)	Size of the quantum dots	D)	Overall thickness of the apparatus
Ans	C)	Size of the quantum dots		

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Department of Chemistry

CHY104--MATERIALS CHEMISTRY FOR DEVICES AND E-WASTE MANAGEMENT

MULTIPLE CHOICE QUESTIONS

UNIT – IV POLYMERS AND NANOMATERIALS

1.	The total number of functional groups or bonding sites present in a monomer molecule is called			
	A)	Degree of Polymerization	B)	Cross-linking
	C)	Functionality	D)	Reactivity
Ans	C)	Functionality		
2.	Bifunctionality required for addition polymerization is provided by			
	A)	Unsaturation present in the monomer	B)	Two functional groups present in monomer
	C)	Catalyst	D)	One functional group present in monomer
Ans	A)	Unsaturation present in the monomer		
3.	 is defined as the number of repeating units in a polymer chain.			
	A)	Functionality	B)	Degree of Polymerization
	C)	Molecular Weight	D)	Monomer
Ans	B)	Degree of Polymerization		
4.	Nylon 6.6 can be prepared by			
	A)	Addition polymerization	B)	Coordination polymerization
	C)	Condensation polymerization	D)	Co-polymerization
Ans	C)	Condensation polymerization		
5.	Vinyl compounds are			
	A)	Bifunctional monomers	B)	Trifunctional monomers
	C)	Polymers	D)	Not example for monomers
Ans	A)	Bifunctional monomers		
6.	The chemical process by which the monomers are converted into polymers is called			
	A)	Initiation	B)	Propagation
	C)	Polymerization	D)	Substitution
Ans	C)	Polymerization		
7.	Which of the following is an example of monomer?			
	A)	Ethane	B)	Ethyl chloride
	C)	Ethene	D)	Ethyl alcohol
Ans	C)	Ethene		
8.	Identify the incorrect statement.			
	A)	Functionality of phenol is 3	B)	Functionality of adipic acid is 2
	C)	Functionality of ethene is 2	D)	Functionality of ethanol is 2
Ans	D)	Functionality of ethanol is 2		
9.	Polymers that soften on heating and harden on cooling are called			
	A)	Thermoplastics	B)	Elastomers
	C)	Thermosetting polymers	D)	Resins
Ans	A)	Thermoplastics		
10.	What is responsible for the plasticity observed in thermoplastics?			
	A)	Strong covalent bonds between polymer chains	B)	Weak secondary forces between polymer chains
	C)	Cross-linking of the polymer chains	D)	Conjugation in the polymer backbone
Ans	B)	Weak secondary forces between polymer chains		
11.	Polymer composites are a blend of			

	A)	Polymer & Metal	B)	Polymer & Fiber
	C)	Polymer & Glass	C)	Polymer & Elastomer
Ans	B)	Polymer & Fiber		
12.	Which material is typically used to bind carbon fibers together in a composite?			
	A)	Glass	B)	Rubber
	C)	Plastic polymer resin	D)	Metal alloy
Ans	C)	Plastic polymer resin		
13.	Which polymer is widely used as the precursor for preparing carbon fiber?			
	A)	Polyethylene	B)	Polyacrylonitrile
	C)	Polystyrene	D)	Polyvinyl chloride
Ans	B)	Polyacrylonitrile		
14.	During the preparation of carbon fiber, at what temperature hydrogen gets expelled to form aromatic rings?			
	A)	200-300 °C	B)	600–700 °C
	C)	1000-1500 °C	D)	2000-2500 °C
Ans.	B)	600-700°C		
15.	Why does undoped polyacetylene behave as an insulator?			
	A)	It has a high density of free electrons	B)	It is not crystalline
	C)	It has a relatively large bandgap	D)	It lacks extensive conjugation
Ans.	C)	It has a relatively large bandgap		
16.	Which of the following acts as a p-type dopant for polyacetylene?			
	A)	Sodium naphthalide	B)	Hydrogen peroxide
	C)	Polyacrylonitrile (PAN)	D)	Iodine (I ₂)
Ans.	D)	Iodine (I ₂)		
17.	In oxidative doping of polyacetylene, species responsible for conduction are			
	A)	Delocalized positive charges	B)	Delocalized negative charges
	C)	Both positive and negative charges	D)	Dopant anions
Ans	A)	Delocalized positive charges		
18.	In reductive doping of polyacetylene, species responsible for conduction are			
	A)	Delocalized positive charges	B)	Delocalized negative charges
	C)	Both positive and negative charges	D)	Dopant anions
Ans	B)	Delocalized negative charges		
19.	Carbon fiber composites are preferred in aerospace because			
	A)	They are cheap and biodegradable	B)	They have high strength and low density
	C)	They have high solubility in solvents	D)	They are elastic
Ans	B)	They have high strength and low density		
20.	With respect to the conductivity observed in doped polyacetylene, which of the following statements holds true for the soliton band?			
	A)	It lies below the valence band.	B)	It is formed only during oxidative doping.
	C)	It disappears upon doping.	D)	It creates energy states within the band gap.
Ans	D)	It creates energy states within the band gap.		

21.	1 nm =			
	A)	10 ⁻⁷ m	B)	10 ⁻⁹ m
	C)	10 ⁹ m	D)	10 ⁻³ m
Ans	B)	10 ⁻⁹ m		
22.	Materials having two dimensions in the nanoscale are called..... nanomaterials			
	A)	Zero Dimensional	B)	One Dimensional
	C)	Two Dimensional	D)	Three Dimensional
Ans	B)	One Dimensional		
23.	An example for zero-dimensional nanomaterials is			

	A)	Nanosheets	B)	Nanotubes
	C)	Nanospheres	D)	Nanowires
Ans	C)	Nanospheres		
24.	An example for one-dimensional nanomaterials is			
	A)	Nanosheets	B)	Nanoplates
	C)	Nanospheres	D)	Nanowires
Ans	D)	Nanowires		
25.	An example for two-dimensional nanomaterials is			
	A)	Nanosheets	B)	Nanotubes
	C)	Nanospheres	D)	Nanowires
Ans	A)	Nanosheets		
26.	Which of the following property is responsible for makes nanomaterials good catalysts?			
	A)	Increased volume	B)	Increase in melting point
	C)	Increase in surface area to volume ratio	D)	Increased conductance
Ans	C)	Increase in surface area to volume ratio		
27.	Photolithography technique for synthesis of nanomaterials makes use of			
	A)	UV radiation	B)	IR radiation
	C)	Visible radiation	D)	Solar radiation
Ans	A)	UV radiation		
28.	Identify the incorrect statement regarding the use of photoresists in photolithography.			
	A)	Negative photoresist leads to the removal of exposed material in the developer.	B)	Positive photoresist dissolves the exposed material in the developer.
	C)	Negative photoresist polymerizes and hardens on exposure to UV light.	D)	Positive photoresist dissolves on exposure to UV light.
Ans	A)	Negative photoresist leads to the removal of exposed material in the developer.		
29.	In photolithography, silica wafers are coated with photoresistor by technique.			
	A)	Dip coating	B)	Spin coating
	C)	Painting	D)	Spray coating
Ans	B)	Spin coating		
30.	Identify the correct sequence of events in photolithography.			
	A)	Cleaning – Development – UV exposure – Photoresist application - Etching	B)	Cleaning – Photoresist application – UV exposure - Development - Etching
	C)	Cleaning – Photoresist application – UV exposure – Etching - Development	D)	Cleaning – UV exposure - Development – Etching – Photoresist application
Ans	B)	Cleaning – Photoresist application – UV exposure - Development - Etching		
31.	Precursor used in sol-gel synthesis of silica is			
	A)	Silicon hydride	B)	Silica
	C)	Silicon alkoxides	D)	Pure Silicon
Ans	C)	Silicon alkoxides		
32.	Choose the INCORRECT statement for the sol-gel method of nanoparticle synthesis.			
	A)	Sol is a stable suspension of colloidal particles in a liquid.	B)	Metal alkoxides are the preferred precursors in sol-gel process.
	C)	The aerogel formation is pressure-driven and not temperature-driven.	D)	The gel is formed by the polymerization reaction.
Ans	C)	The aerogel formation is pressure-driven and not temperature-driven.		
33.	CVD stands for			
	A)	Carbon Vapor Dissolution	B)	Chemical Vapor Deposition
	C)	Carbon Vapor Deposition	D)	Chemical Vapor Dissolution
Ans	B)	Chemical Vapor Deposition		
34.	is the carbon source used in the synthesis of carbon nanotubes by CVD technique.			
	A)	Nitrogen gas	B)	Argon gas
	C)	Hydrocarbon gas	D)	Hydrogen gas
Ans	C)	Hydrocarbon gas		

35.	CVD technique involves			
	A)	Growth of solid structures from solid precursors	B)	Growth of solid structures from gaseous precursors
	C)	Growth of solid structures from solid precursors	D)	Growth of solid structures from molten metal solutions
Ans	B)	Growth of solid structures from gaseous precursors		
36.	What is the purpose of purging hydrogen gas during CNT synthesis via CVD technique?			
	A)	It helps in reducing the catalyst particles and prevents oxidation.	B)	It can convert multi-walled CNT to single-walled CNT.
	C)	It acts as a precursor for CNT formation.	D)	It helps in maintaining the temperature of the reaction chamber.
Ans	A)	It helps in reducing the catalyst particles and prevents oxidation.		

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MULTIPLE CHOICE QUESTIONS

UNIT – V E-WASTE MANAGEMENT

1.	Which of the following is not a source of e-waste?			
	A)	Plastic bottles	B)	Printer
	C)	Fluorescent lamps	D)	Laptops
Ans	A) Plastic bottles			
2.	Choose the incorrect statement.			
	A)	e-waste refers to electronic wastes.	B)	e-waste comprises of only toxic substances.
	C)	It is possible to extract gold from e-waste.	D)	e-waste can cause soil pollution.
Ans	B) e-waste comprises of only toxic substances.			
3.	Recycling of e-waste has a direct impact on			
	A)	Environment	B)	Economy
	C)	Both environment and economy	D)	Neither environment nor economy
Ans	C) Both environment and economy			
4.	Which of the following is an example of halogenated compound found in e-waste?			
	A)	Cd	B)	Toner dust
	C)	CRT	D)	CFC
Ans	D) CFC			
5.	What is the ill-effect of CFC emission?			
	A)	It causes lung cancer.	B)	It results in cardiac failure.
	C)	It obscures vision.	D)	It accumulates in stratosphere and depletes ozone.
Ans	D) It accumulates in stratosphere and depletes ozone.			
6.	The metal ions released from e-waste can prove hazardous to human health due to their			
	A)	easy bioaccumulation	B)	inert nature
	C)	tendency to undergo reduction	D)	optical nature
Ans	A) easy bioaccumulation			
7.	Cadmium present in e-waste mainly affects			
	A)	Kidney and nervous system	B)	Bone and tooth
	C)	Skin	D)	Digestive system
Ans	A Kidney and nervous system			
8.	Which is the first step of e-waste recycling?			
	A)	Chemical dissolution	B)	Physical separation
	C)	Biological treatment	D)	Thermal exposure
Ans	B) Physical separation			
9.	Which of the following technique is not included during pretreatment of e-waste?			
	A)	Mechanical grinding	B)	Treatment with leaching agents
	C)	Sedimentation	D)	Magnetic separation
Ans	B) Treatment with leaching agents			
10.	Magnetic separation of pretreatment enables separation of			
	A)	Noble metals from plastics	B)	Glass wastes from plastics
	C)	Non-ferrous metals from plastics	D)	Ferrous metals from non-ferrous metals

Ans	D)	Ferrous metals from non-ferrous metals		
11.	Direct combustion of e-waste is			
	A)	Safe and environment friendly	B)	Neither safe nor environment friendly
	C)	A high cost-technique	D)	Safe but not environment friendly
Ans	B)	Neither safe nor environment friendly		
12.	Arrange the following processes used during e-waste management in the correct sequence.			
	(i) Separation of metals			
	(ii) Manual dismantling of waste components			
	(iii) Mechanical shredding of the individual components			
	A)	(i), (ii) & (iii)	B)	(ii), (iii) & (i)
	C)	(iii), (i) & (ii)	D)	(ii), (i) & (iii)
Ans	B)	(ii), (iii) & (i)		
13.	Which of the following is not a type of thermal treatment method?			
	A)	Combustion	B)	Sintering
	C)	Extraction	D)	Incineration
Ans	C)	Extraction		
14.	Which of the following method is carried out in the absence of oxygen?			
	A)	Pyrolysis	B)	Gasification
	C)	Combustion	D)	Incineration
Ans	A)	Pyrolysis		
15.	What is the disadvantage of thermal treatment?			
	A)	It is a simple and cheaper method of waste disposal.	B)	It releases large amount of carcinogens in the atmosphere.
	C)	It helps in quick reduction in waste volume.	D)	It is easily adaptable.
Ans	B)	It releases large amount of carcinogens in the atmosphere.		
16.	Pyrometallurgy of e-waste treatment involves action of			
	A)	Pressure	B)	Heat
	C)	Chemicals	D)	Bio-chemicals
Ans	B)	Heat		
17.	During copper smelting process (extraction from ores), the slurry will be rich in			
	A)	Copper	B)	Precious metals such Pt, Sb, etc.
	C)	Sand granules	D)	Plastic particulates
Ans	B)	Precious metals such Pt, Sb, etc.		
18.	In molten salt process, e-waste are treated with			
	A)	Organic solvents	B)	Inorganic salts
	C)	Molten mercury	D)	Organic heterocyclic compounds
Ans	B)	Inorganic salts		
19.	Hydrometallurgy involves treating e-waste with			
	A)	Water	B)	Leaching agents
	C)	Organic solvents	D)	Ionic liquids
Ans	B)	Leaching agents		
20.	Most preferred leaching agent for recovery of lithium is			
	A)	Cyanides	B)	Thiosuphate
	C)	Acid	D)	Thio urea
Ans	C)	Acid		
21.	Which are the two stages of hydrometallurgy?			
	A)	Leaching and electrodeposition	B)	Electrodeposition and recover
	C)	Leaching and recovery	D)	Extraction and leaching
Ans	C)	Leaching and recovery		
22.	Which of the following method is not a part of pyrometallurgical treatment of e-wastes?			
	A)	Smelting	B)	Incineration
	C)	Pyrolysis	D)	Filtration
Ans	D)	Filtration		

23.	Identify the incorrect statement.			
	A)	Pyrometallurgy involves high temperature treatment.	B)	Combustion is reduction at room temperature.
	C)	Leaching involves conversion of metals into soluble salts or complexes.	D)	Pyrolysis is undertaken in the absence of oxygen.
Ans	B)	Combustion is reduction at room temperature.		
24.	Extraction of metals by treatment with HCl is called as leaching.			
	A)	Cyanide	B)	thioruea
	C)	Acid	D)	thiosulphate
Ans	C)	Acid		
25.	Which of the following method is not suitable for gold extraction?			
	A)	Cyanide leaching	B)	Metal leaching
	C)	Halide leaching	D)	Thiosulphate leaching
Ans	B)	Metal leaching		
26.	Which leaching agent is commonly employed for gold recovery from e-waste?			
	A)	Aqua regia or cyanides	B)	Strong ammoniacal solution
	C)	Mild organic acids	D)	Mixture of ionic salt and weak base
Ans	A)	Aqua regia or cyanides		
27.	During thiourea leaching, reaction takes place between the metal ion and thiourea.			
	A)	Repulsion	B)	Neutralization
	C)	Substitution	D)	Complexation
Ans	D)	Complexation		
28.	 is a metal recovery process consisting of transfer of metal ions from inorganic to organic phase.			
	A)	Solvent Extraction	B)	Adsorption
	C)	Ion exchange	D)	Electrodeposition
Ans	A)	Solvent Extraction		
29.	Which of the following statement is not true for ion exchange process?			
	A)	It is an extension of solvent extraction method.	B)	The exchange reaction occurs on the surface of polymer beads.
	C)	The method is not selective.	D)	It is an environment-friendly method.
Ans	C)	The method is not selective.		
30.	In solvent extraction of metal recovery of e-waste, leached solutions are treated with			
	A)	Excess water	B)	Concentrated acids
	C)	Concentrated alkalis	D)	Organic solvents
Ans	D)	Organic solvents		
31.	Utilization of activated charcoal is effective for process.			
	A)	solvent Extraction	B)	adsorption
	C)	ion exchange	D)	electrodeposition
Ans	B)	Adsorption		
32.	In the ion exchange process used for metal recovery, which of the following is true?			
	A)	Not environment friendly	C)	Needs high concentration of ions
	C)	Highly selective	D)	Needs high temperature
Ans	B)	Highly selective		
33.	Chose the correct statement.			
	A)	Metal recovered by electrodeposition method is of high purity.	B)	During ion exchange, the metal ions bind to the polymeric functional groups by hydrogen bonding.
	C)	Solvent extraction is a three-phase system.	D)	Activated charcoal has the least tendency to behave as metal adsorbent.
Ans	A)	Metal recovered by electrodeposition method is of high purity.		
34.	 metal is taken as cathode during gold recovery by electro-deposition.			
	A)	Pt	B)	Cu
	C)	Au	D)	Co
Ans	C)	Au		

35.	During the electrodeposition process in e-waste recovery, the recovered metal			
	A)	Gets released from the anode	B)	Gets deposited on the cathode
	C)	Increases the potential of the process	D)	Gets converted to its metal oxide
Ans	B)	Gets deposited on the cathode		
36.	CPCB and SPCB are part of			
	A)	Producer assets	B)	Consumer groups
	C)	Recycling companies	D)	Statutory government regulatory bodies
Ans	D)	Statutory government regulatory bodies		
37.	As a stakeholder for e-waste management, which of the following is not the responsibility of a recycler?			
	A)	Collection of e-waste	B)	Registering themselves with SPCB
	C)	Formulation of legal frameworks	D)	Following the standard norms of recycling
Ans	C)	Formulation of legal frameworks		
38.	Who, among the following, frames the guidelines for e-waste disposal and recycling?			
	A)	CPCB	B)	Producers
	C)	Consumers	D)	Recyclers
Ans	A)	CPCB		
39.	A responsible consumer must not			
	A)	Segregate the e-waste from regular wastes	B)	Frequently purchase excessive electronic devices
	C)	Be aware of the regulations on e-waste disposal	D)	Encourage others to recycle e-wastes scientifically
Ans	B)	Frequently purchase excessive electronic devices		
